CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester Bachelor of Technology (CE)

University Examination May 2011

CE209: Design and Analysis of Algorithm

Date: 13/05/2011 **Time: 10:00 a.m to 1:00 p.m** **Maximum Marks:70**

Instructions:

1. *Attempt all questions.*
2. *Answers both sections in separate answer books.*
3. *Figure to the right indicates full marks.*
4. *Assume suitable data, if necessary and mention it.*

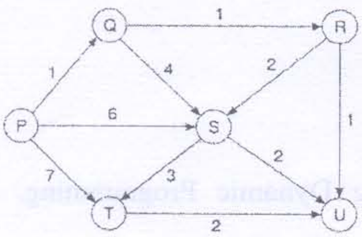
SECTION-I

Q-1

[A]	What is an algorithm? List out the criteria that all algorithms must satisfy.	2
[B]	What is Order of Growth? Explain with the help of an example.	2
[C]	Explain Divide and Conquer technique and give the general divide and conquer recurrence.	2
[D]	Define a recursive algorithm to compute the n^{th} Fibonacci number and analyze its efficiency.	2
[E]	Define the time complexity and space complexity of an algorithm. Differentiate between average case, best case and worst case time complexities.	3

Q-2

[A] Dijkstra's single source shortest path algorithm on the following edge weighted directed graph with vertex P as the source is given below. **6**



1. Output the sequence of vertices identified by the Dijkstra's algorithm for single source shortest path when the algorithm is started at node P.
2. Write down the sequence of vertices in the shortest path from P to U. What is the cost of shortest path from P to U?

[B] The array given as input to a quick sort algorithm is already sorted. What will be the running time of quick sort if the pivot is taken to be (i) the first element (ii) the last element and (iii) median of three(first, middle, last) elements ? Give suitable example to support your answer. **4**

[C] Write a recursive algorithm for Binary Search. What is the time complexity of the algorithm? What is the size of each partition? **2**

OR

Q-2

[A] Simulate both the insertion sort and the selection sort algorithms on the following two arrays: U =[1,2,3,4,5,6] and V = [6,5,4,3,2,1].Does insertion sorting run faster on the array U or the array V? or selection sorting? Justify your answers. **6**

[B] Differentiate between brute force approach and backtracking. **3**

[C] Specify the algorithms used for constructing Minimum cost spanning tree. Which algorithm is preferable if the given graph is **3**
1. Dense Graph 2. Sparse Graph

Q-3 Attempt Any Three **12**

[A] Write krushkal's algorithm for finding minimum spanning tree of a connected weighted graph. Show using example that the algorithm you have written indeed determines the optimal spanning tree. On what approach this algorithm is based on?

[B] What is Asymptotic analysis? Explain the various asymptotic notations used in algorithm design.

[C] Define Following Terms:
1) Connected Component 2) Degree of a Graph 3) Articulation Point 4) Spanning Tree

[D] Explain a Generic Greedy Algorithm and functions used in Greedy approach.

[E] What is n-Queen's problem? Generate the state space tree for n = 4.Write all possible feasible solutions.

SECTION-II

Q-4

[A] Dynamic Programming is bottom up problem solving approach. Justify 2

[B] List down the factors which affect the efficiency of backtracking algorithm. 2

[C] State Master's Theorem. 2

[D] What is a feasible solution and what is an optimal solution in greedy technique? Give suitable example. 2

[E] What is the basic idea behind String Matching? Write Naive Algorithm for String Matching. 3

Q-5

[A] Solve the following recurrence equations. 5
$A_n=A_{n-1}+2A_{n-2}$ initial condition $A_0=2, A_1=7$.

[B] Define Principle of Optimality. Explain how it holds on the Shortest Path Problem. 2

[C] Use branch and bound to solve the assignment problem (Minimization) with the following cost matrices. 5

Job	1	2	3	4
Person a	58	41	11	24
Person b	9	44	21	23
Person c	46	20	32	11
Person d	12	10	15	16

OR

Q-5

[A] Explain topological sort with example. 5

[B] Find the optimal sequence of Matrix Chain Multiplication using Dynamic Programming. 4
A=20x15, B=15x30, C=30x10 D=10x2.

[C] Critically differentiate greedy strategy and dynamic programming approach with the help of two points. 3

Q-6 Attempt Any Three 12

[A] Write mathematical formulation of 0-1 knapsack problem. Use dynamic programming approach to solve the following instance of the problem
Maximum capacity = 11 units
No of items = 5
Weights = 1,2,5,6,7
Profits = 1,6,18,22,28

[B] Find the longest common subsequence between strings: ALGORITHM & LOGARITHM.

[C] Suppose that you are a cashier in a strange country where the currency denominations are: 1, 3, 8, 16, 22, 57, 103, and 526 cents (or more generally d0, d1... dn-1). Describe a dynamic programming algorithm to make change for C cents using the fewest number of coins.

[D] Define P, NP and NP-Complete problems.

[E] Write Floyd's Algorithm for finding All Pairs of Shortest Path.